PROJECT INTRODUCTION · 30 SEPTEMBER 2026

Exoplanet Hunter V2

A machine-learning system that looks at signals from NASA's planet-hunting telescopes and estimates which ones are real planets. What it is, why I built it, how it works end to end, where it stands, and what comes next.

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In one paragraph

NASA's TESS and Kepler telescopes watch hundreds of thousands of stars for tiny, regular dips in brightness: the shadow of a planet crossing its star. Their pipelines flag thousands of such signals, and many of the flags are not planets. They are pairs of stars eclipsing each other, a neighbouring star's light blended in, or instrument noise. Confirming a planet costs scarce telescope time. **Exoplanet Hunter takes each flagged signal, reads its light curve and a few facts about its star, and returns a calibrated probability that it is a planet**, so the follow-up budget goes to the most promising candidates. It runs live as a scoring API and a web console.

Where it is going. Vetting known signals comes first. The destination is **discovery**: searching raw TESS light curves for signals nobody has flagged, and using the vetter to decide which are worth a telescope's time. That only makes sense once the vetter is accurate, which is why the current work is about measuring and fixing it.

At a glance

Task	Binary classification of transit signals: planet or false positive
Data	5,826 labelled stars from TESS, Kepler and K2; 5,104 unresolved TESS candidates to rank
Inputs	Two phase-folded light-curve views, plus 9 numbers about the signal and the star
Model	Dual-view 1-D convolutional network with attention, Platt-calibrated, 5-fold cross-validated
Served result	TESS ROC-AUC 0.910; 30.7% of TESS planets kept at a 1% false-positive rate (Kepler: 0.991, 79.9%)
Method	Model comparisons read against a measured noise floor; the main experiments pre-registered
Status	Live. An audit in September 2026 found input defects and a missing baseline; fixing them before retraining
Stack	Python, TensorFlow/Keras, lightkurve, scikit-learn, FastAPI, DVC, MLflow, Prefect

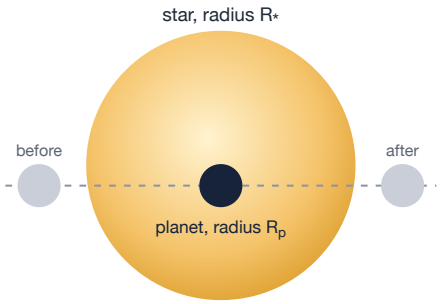
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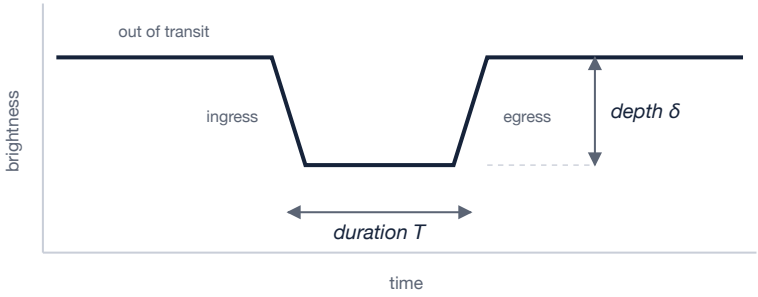
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THE SCIENCE

How a planet shows up in a light curve

A transiting planet blocks a little of its star's light once per orbit. A light curve is the star's brightness over time; the job is to decide whether a regular dip in it was made by a planet.





$\delta \approx (R_p / R_*)^2$ depth: the fraction of the star's disc covered

$a^3 = G\,M_*\,P^2 / 4\pi^2 \;\Rightarrow\; a/R_* = (G\,\rho_*\,P^2 / 3\pi)^{1/3}$

Kepler's third law

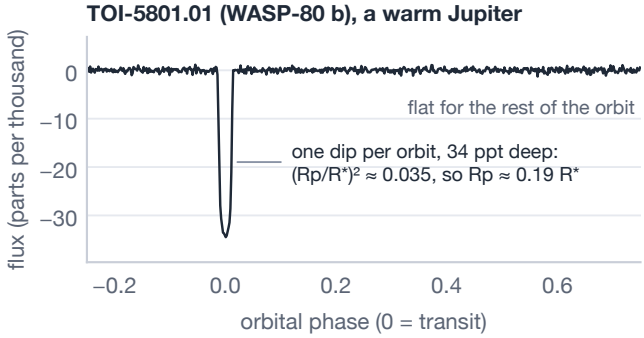
$T \approx (P/\pi) \cdot \arcsin(R_*/a)$ duration of a central transit, circular orbit

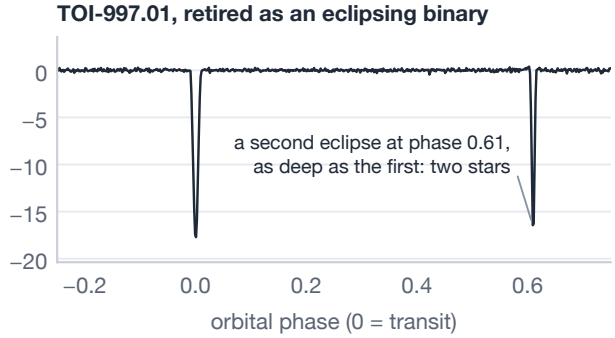
$S/N = (\delta / \sigma) \cdot \sqrt{N_{tr}}$ signal-to-noise over N_{tr} transits

What the numbers tell you. A 1% dip on a Sun-like star means a planet about Jupiter's size; an Earth makes a dip of 0.008%. The period *P* is the time between dips. The star's density sets how long a planet on that orbit can take to cross, so a duration inconsistent with the star is a warning sign. Averaging many transits beats down the noise by $\sqrt{N_{tr}}$, which is why the signal is **phase-folded**: every transit is stacked on top of the others.

Why many flags are not planets.

- **Eclipsing binaries:** two stars eclipsing each other. Tell-tales are a **secondary eclipse**, usually near half an orbit later, and odd and even eclipses of different depths.
- **Blends:** an eclipsing binary on a nearby star, diluted into the target's pixels. The centre of the light can shift during the dip.
- **Artefacts and variability:** spacecraft momentum dumps, scattered light, or the star's own spots and pulsations mimicking a dip.





Two real TESS targets from the labelled set, cleaned, detrended and folded over their whole orbits by the project's own code. Left: TOI-5801.01, the known warm Jupiter WASP-80 b (a 3.4% dip on a small star; its published radius is 0.95 Jupiter radii, a little under the rough depth estimate, because limb darkening deepens the middle of a transit). Right: a signal retired as an eclipsing binary, whose second, equally deep eclipse gives it away. The served model scores the planet 0.99 and the binary 0.05.

WHY IT EXISTS

A coursework project pointed at a real problem

I started this to use what my degree was teaching on data with real consequences. It is on its second version: the first proved a model could be trained; the second rebuilt everything around measuring whether the model is actually right.

What the courses contributed

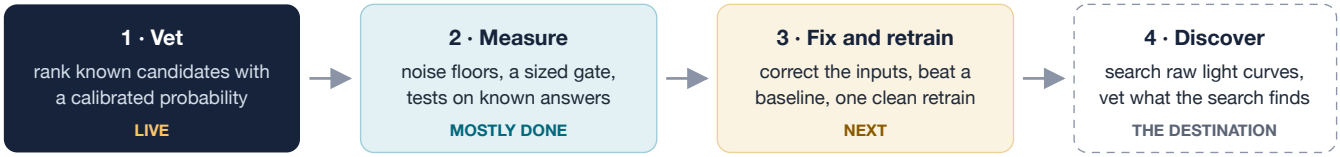
COURSE	WHERE IT SHOWS UP
DATA 305	The convolutional network in Keras, the data pipeline (lecture 6) and the training loop (lectures 4 and 5)
DATA 303	Cross-validated evaluation and model comparison (§7.1–7.2)
STAT 292	Paired and blocked comparisons: each star as its own control in the injection tests
STAT 293	ANOVA and the design of experiments: the injection grid as a blocked factorial; error terms for noise floors
DATA 301	The data-science workflow, and writing the work up as a report
AIML 131, 231	The machine-learning foundations everything above builds on

Géron's *Hands-On Machine Learning* has a project checklist that says to "*train many quick-and-dirty models from different categories*" before settling on one. Doing that in September 2026 is what exposed the problems on page 13.

Version 1 to version 2

- **V1** (first half of 2026): a dual-view CNN with attention, MLflow tracking and Hydra configs. It trained, and it scored well on a pooled number.
- **V2** (July to now): a clean rebuild. Data validation gates that stop a run rather than warn, a calibrated cross-validated ensemble, a **promotion gate** that decides whether a new model may replace the served one, a public API and console, a written report, and a frozen record of its experiments.
- The largest single effort was an eleven-branch network inspired by NASA's ExoMiner. It was built, measured against the gate and closed: it did not beat the served model on the metric that mattered then.

What it aims to do



Why discovery waits. A blind search of TESS light curves turns up far more periodic signals than planets, most of them noise or binaries. The vetter is the filter between that flood and a telescope. If the vetter cannot see a secondary eclipse, or does not respond to a faint transit, a discovery run would hand astronomers a confident list of mostly false positives. So the order is: make the vetter measurably accurate, then point it at the unknown. The search code already exists (a box-least-squares search); what is deferred is running it at survey scale.

> **The rule the project is built around:** a difference between two models that is smaller than the noise you would see retraining the same model with a different random seed is not a result. The gate reads model comparisons against that noise floor, and for the main experiments, how a result will be read is written down before it is run.

THE WHOLE PIPELINE

From NASA archives to a ranked shortlist

Read left to right, then down. Each box is a stage in the code; the red box marks the defect found in September 2026; the dashed row is where the project is headed.



DATA

Where the labels come from, and what they mean

A supervised model is only as good as its labels. Here a label is an astronomer's final verdict on a signal, taken from public archives.

Sources

SOURCE	WHAT IT PROVIDES
NASA Exoplanet Archive	TESS Objects of Interest (TOI), Kepler Objects of Interest (KOI, DR25), K2 candidates: verdicts, periods, epochs, depths
ExoFOP	The TESS follow-up team's verdicts and comments
MAST	Light curves from the missions' own pipelines (SPOC for TESS)
Gaia DR3, TIC	Stellar temperature, radius, surface gravity, brightness

Label rules

ARCHIVE VERDICT	LABEL
Confirmed or known planet (CP, KP, CONFIRMED)	**1**, planet
False positive, false alarm, refuted (FP, FA, REFUTED)	**0**, not a planet
Still a candidate (PC, APC, CANDIDATE)	held out: never trained or evaluated on; these are what the model exists to rank

The labelled set

MISSION	FALSE POS.	PLANETS	TOTAL
TESS	1,373	1,423	2,796
Kepler (capped)	1,250	1,250	2,500
K2	215	315	530
All	**2,838**	**2,988**	**5,826**

Committed on `main` at the 13 September 2026 refresh. The served model was trained in July on an earlier table and has never seen K2. The candidate table holds 12,472 TESS signals (TOI 8,148, community TOI 4,324), resolved ones included for reference; 5,104 are genuinely unresolved, and the live service has scored about 3,900 of those.

Design choices, and what each costs

- **One row per host star.** No star can land in two cross-validation folds, so no star is both trained and tested on. The cost: every extra planet in a multi-planet system is dropped.
- **Kepler is capped at 1,250 of each class**, a fixed hashed sample for balance. It is 44% of the 5,664 eligible Kepler rows.
- **What "false positive" means differs by mission.** Kepler negatives are false positives from DR25, the Kepler Robovetter's single uniform automated run (score below 0.5), and its positives are confirmed planets. TESS negatives are signals that passed initial vetting to become TOIs and were ruled out only after follow-up.
- **Derived data is versioned** with DVC, so a result can be traced back to the exact data it came from.

A confound the data carries. On TESS, how long a star has been observed correlates with its label (Spearman +0.387): confirmed planets have a median observing baseline of 1,495 days against 430 for false positives, because planets attract more observation. A model can gain apparent skill by learning observing history rather than astrophysics. The served model's score correlates with baseline at +0.381. It inherits the confound but does not amplify it. An earlier branch model amplified it to +0.516.

PREPROCESSING

From a raw light curve to what the model sees

Four steps turn years of noisy photometry into two fixed-length arrays. The design follows Shallue & Vanderburg (2018), one of the first papers to apply deep learning to transit vetting.

1 · Clean. Drop missing points and clip bright outliers (cosmic rays, pointing jumps) above 5σ. Only the upper tail is clipped, because a symmetric clip would delete the very dips being studied.

2 · Detrend. Stars vary on their own (spots, pulsation). A Savitzky–Golay filter (window 301 cadences, polynomial order 2) is fitted and divided out, with the in-transit points *masked* from the fit, so the filter cannot learn to flatten the transit.

3 · Fold. Every time stamp is mapped to orbital phase, so all transits land on top of each other:

$$\varphi = ((t - t_0) / P + ½) \bmod 1 - ½$$ φ = 0 at mid-transit

4 · Bin and normalise. Median-bin into a **global view** (2,001 bins, meant to cover the whole orbit) and a **local view** (201 bins, meant to cover ±3 transit durations), then scale so the baseline is 0 and the deepest bin is −1:

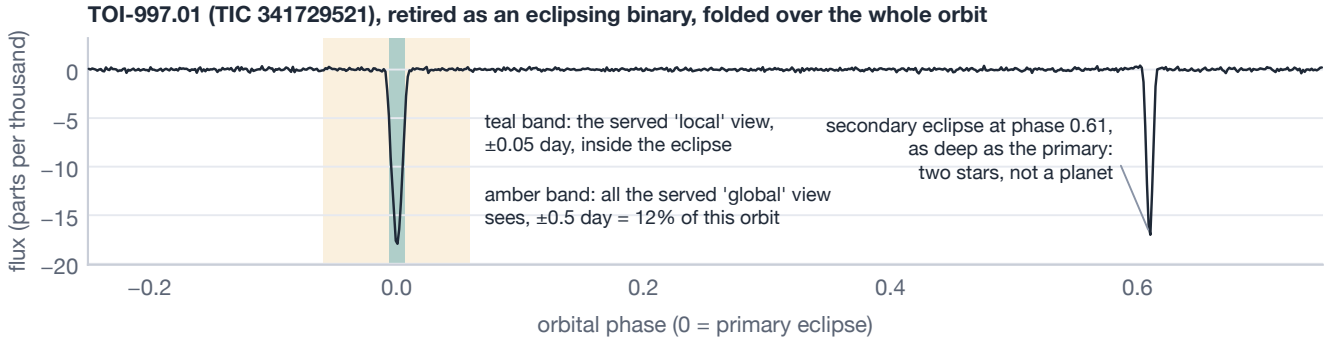
$$v' = (v - \text{med } v) / |\min(v - \text{med } v)|$$ shape, not depth; depth is given separately

```python
# preprocess/views.py (served code, abridged)
def build_views(lc, period, t0, duration, *,
                global_bins=2001, local_bins=201,
                local_durations=3.0):
    # meant as phase units
    lo, hi = -0.5, 0.5
    half = local_durations * duration / period
    _, g = fold_and_bin(lc, period, t0,
                        global_bins, lo, hi)
    _, l = fold_and_bin(lc, period, t0,
                        local_bins, -half, half)
    return Views(_normalise(g), _normalise(l))

# preprocess/fold.py (abridged)
def fold_to_profile(lc, period, t0, n, lo, hi):
    folded = lc.fold(period=period, epoch_time=t0)
    # folded.time is in DAYS, so lo and hi are days
    return bin_profile(folded.time.value,
                       folded.flux.value, n, lo, hi)

# the fix (issue #140): fold in phase
    folded = lc.fold(period=period, epoch_time=t0,
                     normalize_phase=True)
```

The defect. The windows are written in phase units but applied to a fold measured in days. So the "global" view spans ±0.5 day, which is 10% of a 10-day orbit, and never reaches phase 0.5, where a secondary eclipse usually sits, for 92% of TESS targets. The "local" view spans ±3/P transit durations, which falls entirely inside the transit for periods over 6 days (29% of TESS targets). It was found by an audit on 14 September and is fixed with the next retrain, because changing the views changes every input the served model was trained on.



What the model was meant to see, and what it sees. A real eclipsing binary folded over its whole 8.4-day orbit. The shaded bands are the only parts the served views contain: 12% of the orbit for the "global" view, and a sliver inside the eclipse for the "local" view. The secondary eclipse, the clearest evidence that this is two stars, is never shown to the network. It still scores this binary 0.05 (page 2), and the evidence points to the primary eclipse's shape, which the window does contain: V-shaped, only 0.53 of its full duration wide at half depth, against medians of 0.82 for TESS planets and 0.61 for false positives among signals as strong as this one (S/N above 100). Flattening the window raises its score to 0.41.

FEATURES

What the model is told about each signal

Alongside the two views, the network gets a short vector of numbers: facts about the star, the catalogue's measurements of the signal, and diagnostics that astronomers use to catch false positives. The served model uses the first nine; the current build computes thirteen.

#	FEATURE	DEFINITION	WHY IT HELPS	SERVED
0–3	**Star**	effective temperature, radius, surface gravity (log g), TESS magnitude	A planet's size and orbit have to be plausible for its star	yes
4–6	**Signal**	catalogue depth δ, duration *T*, log period	Binaries tend to be deeper and shorter-period	yes
7	**Signal-to-noise**	catalogue S/N (served); pink-noise S/N from the light curve (13-number build)	How far above the noise the dip is	yes
8	**Centroid shift**	in-transit minus out-of-transit centre of light, per axis, in units of its scatter, added in quadrature	If the light's centre moves during the dip, the dip is on a neighbouring star	yes
9	**Odd/even depth**	difference between odd- and even-numbered transit depths, in standard errors	An eclipsing binary at half its true period alternates deep and shallow eclipses	—
10	**Odd/even timing**	difference in odd and even transit mid-times (Kunimoto 2025, eq. 13)	Catches eccentric binaries whose eclipses are not half an orbit apart	—
11	**Secondary eclipse**	significance of the deepest duration-wide dip outside ±2 durations of the transit (Kunimoto 2025, §3.9)	A planet's own eclipse is tiny; a star's is not	—
12	**Duration check**	observed *T*/*P* against a central, circular transit for this star's density (below)	The most effective false-alarm test in Kunimoto 2025	—

The maths behind two of them

$$\mathrm{S/N} = \delta / \sigma_{\mathrm{tr}}, \quad \sigma_{\mathrm{tr}} = \sqrt{(\sigma_w^2 / n + \sigma_r^2 / N_{\mathrm{tr}})}$$ pink noise (Pont 2006)

σ_w is the white (point-to-point) noise over n in-transit points; σ_r is the red, correlated noise measured by binning the light curve at the transit duration. Correlated noise does not average down with more points, only with more transits.

$$q_{\mathrm{circ}} = (1/\pi) \cdot \arcsin(R_* / a), \quad \text{flag if } q_{\mathrm{circ}}/q < 0.6,\ q > 0.5 \text{ or } a/R_* < 1.5$$

Here $q = T/P$, and a/R_* comes from Kepler's third law and the star's density (page 2). A central transit is the longest a circular orbit allows, so one much longer is hard to explain with a planet. The model is given the ratio itself, not the flag. The console's warning flag currently tests the ratio the other way up, catching transits that are too short; that is being fixed (#152).

Handling the numbers

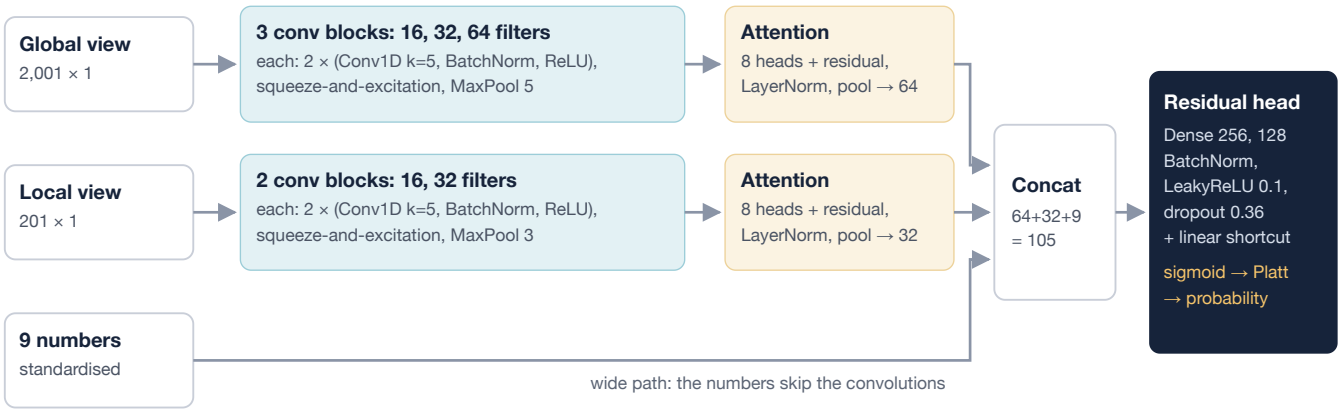
- **Missing values** (no Gaia match, no centroid data) are imputed with the training fold's median.
- **Heavy tails** (centroid S/N, secondary significance) are log-compressed, signed where values can be negative, before standardising.
- The imputer and scaler are **fitted per fold on training rows only** and saved with the model, so scoring replays exactly the training transform.

Why the extra four matter. They carry, as numbers, evidence the served views do not show: secondary eclipses fall outside the defective windows, and folding averages odd and even transits together. In an exploratory check, a simple model given all thirteen scored higher than the served network on TESS (page 13).

THE MODEL

A dual-view convolutional network

Two one-dimensional convolutional towers read the two views, each finishing with self-attention; the nine numbers join them in a "wide" path; a residual head combines everything into one probability.



The pieces, and why each is there

- **Convolutions** learn local shapes: the ingress, the flat or V-shaped bottom, a bump.
- **Squeeze-and-excitation** (Hu 2018) lets each block re-weight its own channels:

$$s = \sigma(\, W_2\, \mathrm{ReLU}(\, W_1 \cdot \mathrm{mean}_t(x)\,)\,), \quad x' = s \odot x$$

- **Multi-head self-attention** lets any part of a view attend to any other, so the network can relate the transit to features far from it; a residual connection and LayerNorm keep it stable: $h = \mathrm{LN}(F + \mathrm{MHA}(F, F, F))$.
- **The wide path** feeds the nine numbers straight into the head, where they meet the learned light-curve summaries.
- **Dropout stays on at inference** (Monte Carlo dropout), so repeated passes give a spread, not just a point.

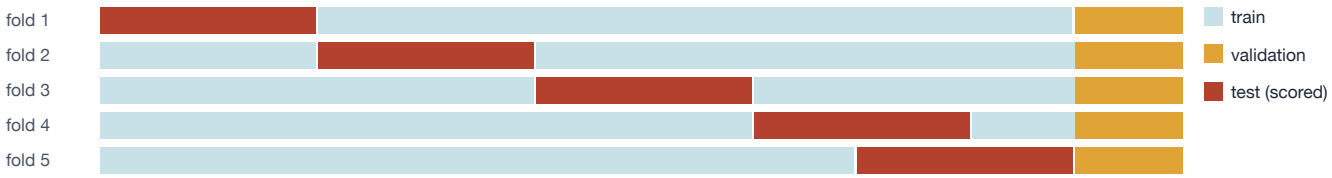
```python
# models/cnn_dualview.py (abridged)
def _se_block(x, *, reduction=8, floor=4):
    c = int(x.shape[-1])
    s = layers.GlobalAveragePooling1D()(x)
    s = layers.Dense(max(c // reduction, floor),
                     activation="relu")(s)
    s = layers.Dense(c, activation="sigmoid")(s)
    s = layers.Reshape((1, c))(s)
    return layers.Multiply()([x, s])

g = _conv_tower(g_in, [16, 32, 64],
                conv_per_block=2,
                kernel_size=5, pool_size=5)
g = _attention_pool(g, num_heads=8, key_dim=32)
l = _conv_tower(l_in, [16, 32],
                conv_per_block=2,
                kernel_size=5, pool_size=3)
l = _attention_pool(l, num_heads=8, key_dim=32)
z = layers.Concatenate()([g, l, aux_in])  # wide
h = _residual_fusion_head(z, fc_units=[256, 128],
                          dropout=0.36)
out = layers.Dense(1, activation="sigmoid")(h)
```

Lineage: the two-view design is Shallue & Vanderburg's AstroNet; squeeze-and-excitation placement follows Xie et al. 2025; attention on each tower follows ExoNet-style work; the residual late-fusion head keeps the fusion path learning before the towers converge.

How the model is fitted, and why its outputs are probabilities

Five-fold cross-validation gives every labelled star exactly one prediction from a model that never trained on it. That "out-of-fold" set is what the performance numbers in this document are computed on.



Each star is in exactly one test fold. Within each training fold, 12% is held back as validation: it decides when to stop training and fits the calibrator. (Schematic: the validation slice is drawn at the end for clarity; it is a stratified random subset.)

Fitting

$$\mathcal{L} = -(1/N)\,\Sigma\, w_{y_i}\,[\, y_i \log p_i + (1 - y_i)\,\log(1 - p_i)\,]$$

class-weighted cross-entropy

- Adam, learning rate 3.2×10^{-4}, gradient clipping at 1.0, batch size 32, L2 regularisation 5×10^{-5}.
- Up to 200 epochs; **early stopping** when validation ROC-AUC has not improved for 25 epochs, restoring the best weights; learning rate halved on plateaus.
- Hyperparameters from a 33-trial Optuna search (July 2026); the optimum was flat near the defaults.
- Augmentation, fresh every epoch: small phase shifts, added noise, ±5% amplitude scaling, random masking of 2% of bins. The served run trains one model per fold; the plan moves to five (page 15).

Calibration: making 0.9 mean nine in ten

A network's raw sigmoid output ranks well but is not a probability. **Platt scaling** fits a two-parameter logistic map on each fold's validation scores:

$$p = \sigma(\, a \cdot s + b\,)$$ fitted by logistic regression on validation data only

It is affine, so unlike temperature scaling it can also move the decision boundary. Within a fold neither changes the ranking. Across folds Platt scaling does, because each fold gets its own map: putting all five on one probability scale lifts the pooled TESS ROC-AUC from 0.904 to 0.910.

Uncertainty

$$\bar{p} = (1/T)\,\Sigma_t\, p_t, \quad \sigma^2 = \sigma_{\mathrm{MC}}{}^2 + \sigma_{\mathrm{fold}}{}^2$$ MC dropout plus fold disagreement

The console shows the spread over dropout passes and the disagreement between the five fold models separately, because they measure different things: how unsure one network is, and how much the choice of training data matters.

EVALUATION

How it is scored, and how it scores

The model's job is to rank a shortlist for a fixed telescope budget, so the headline is not accuracy. It is how many real planets survive a very strict cut.

$$\mathrm{AUC} = \mathrm{P}(s^{+} > s^{-})$$ chance a random planet outscores a random false positive

$$\text{recall@1\%} = \max \mathrm{TPR} \text{ subject to } \mathrm{FPR} \le 0.01$$ the shortlist operating point

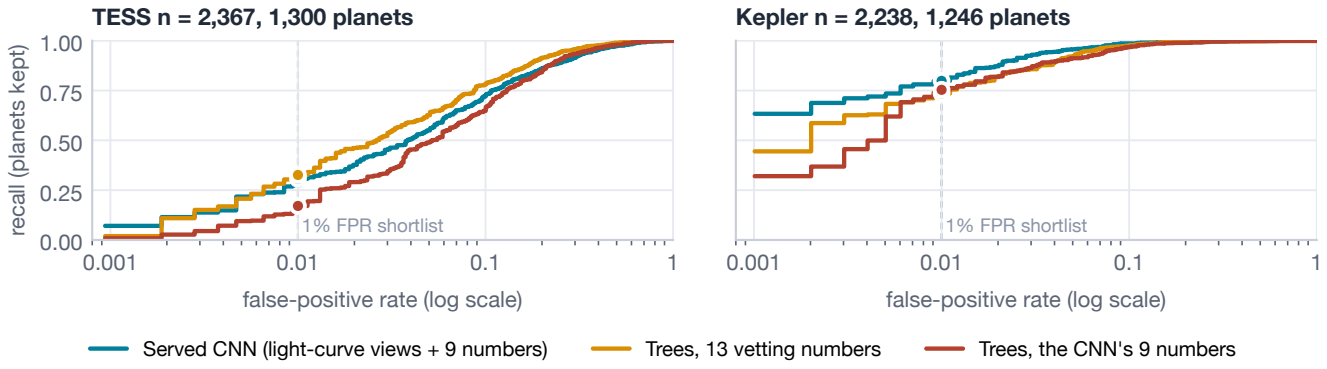
$$\mathrm{Brier} = (1/N)\,\Sigma\,(p_i - y_i)^2$$ mean squared error of the probabilities

$$\mathrm{ECE} = \Sigma_b\,(n_b/N)\,|\,\bar{y}_b - \bar{p}_b\,|$$ calibration gap over score bins

Why not accuracy? The labelled set is balanced by construction, but the candidate pool is not, and its true planet fraction is unknown (the model itself puts 83% of scored candidates above 0.5). Accuracy depends on that base rate; ranking metrics do not. **Why TESS decides?** TESS is the mission still finding new candidates, and it is the harder one.

OUT OF FOLD, SERVED RUN	TESS	KEPLER
Stars (planets)	2,367 (1,300)	2,238 (1,246)
ROC-AUC	0.910 0.899–0.922	0.991 0.989–0.994
Recall at 1% FPR	**0.307** 0.222–0.361	**0.799** 0.742–0.875
Brier score	0.121	0.036
Expected calibration error	0.044	0.041

95% bootstrap intervals. The recall interval is wide because the 1% cut sits at the tenth-highest-scoring false positive: about ten rows decide it. Counts are the served run's evaluation population, the label table as it stood on 17 August 2026; the current table (page 5) is larger and adds K2, which this run never saw. The live console and `docs/report.pdf` show 31.1% (Kepler 81.3%) for the same model: they place the cut by score quantile, which lets one more false positive through (1.03%). Issue #144 unifies the two.



Out-of-fold ROC curves on a logarithmic false-positive axis, so the 1% shortlist region is visible (dots mark it). Teal is the served network. Amber and red are simple tree-model baselines from the independent check on page 13, drawn here for scale.

The Kepler–TESS gap is the first thing to explain. The same model keeps 80% of Kepler planets at a 1% false-positive rate but only 31% of TESS planets. Part of that is the data: Kepler stared at one field for four years, while TESS gets 27 days per sector. Part may be the labels: Kepler's false positives come from one uniform automated vetting run, TESS's from mixed human follow-up. A tree model given no light-curve views at all shows the same gap (AUC 0.985 against 0.904). The two explanations have not yet been separated, and the one label effect tested so far, the Robovetter score cut, accounts for less than 0.01 of the gap.

Noise floors and the promotion gate

Train the same network twice with different random seeds and its TESS AUC moves. A challenger that "wins" by less than that is not better. The promotion gate turns this into a rule, and in September 2026 the rule itself was measured and repaired.

The rule

$$\mathrm{se}(\Delta) = \sqrt{\left(\sigma^2/n_c + \sigma^2/n_i + \sigma_{\mathrm{samp}}{}^2\right)}$$

σ is the seed-to-seed spread of one model's TESS AUC; n_c, n_i the number of models per fold for the challenger and the incumbent; σ_{samp} the sampling noise of the rows (0.0021).

$$\mathrm{floor} = 2 \cdot \mathrm{se}(\Delta)$$

$$\mathrm{MDE} = (z_{0.975} + z_{0.8}) \cdot \mathrm{se} = 2.802 \cdot \mathrm{se}$$

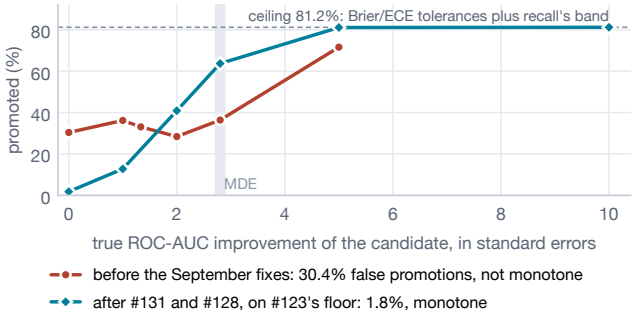
MARGIN Δ IN TESS AUC	VERDICT
Δ > floor, calibration and recall not worse	**PROMOTE** (a person still decides)
0 < Δ ≤ floor	**UNRESOLVED**: the instrument cannot tell
Δ ≤ 0, or a guard fails	**REJECT**

The minimum detectable effect (MDE) is the smallest real improvement an AUC-only test would catch 80% of the time. With five models per side it is about 0.0126 AUC; against the current single-model champion it is 0.0199. The full gate catches less: about 64% at the MDE with five models per side (chart), because its Brier and ECE tolerances cap it (#134).

Estimating σ from a few models

$$\hat{\sigma} = \sqrt{\left((1/F)\,\Sigma_f s_f^2\right) / c_4\left(F(M-1) + 1\right)}$$

The variance of M models' AUCs within each of F folds, pooled, then bias-corrected with $c_4(n) = \sqrt{(2/(n-1))}\,\Gamma(n/2)/\Gamma((n-1)/2)$. Merged in September 2026 (PR #139); it is measured on the TESS rows the gate actually decides on.



The gate simulated 6,000 times per point (five models against five). The value at zero is how often it promotes a candidate that is truly no better. Before September's fixes that was 30%, and a better candidate could be promoted less often than a worse one. After them it is about 2% (1.8% with the exact noise term plotted here, 2.1% as merged), close to the 2.3% a two-standard-error rule should give, and the curve is monotone.

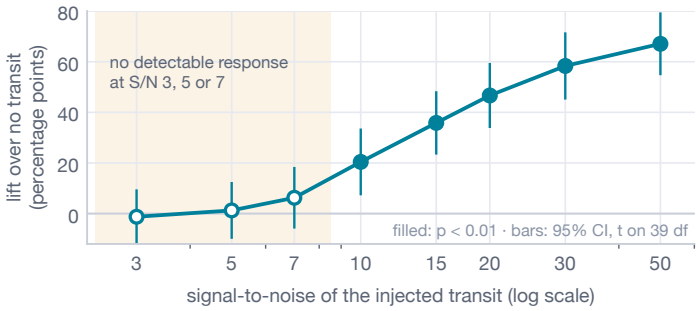
Pre-registration. Before a pre-registered experiment runs, a file says what will be measured, what result would count as success, and what would count as failure. It is committed before the result exists and not edited afterwards. In September 2026, four predictions were falsified and recorded as such, including one fix that passed its main test but cut the gate's power to 1.6%.

What is still open: fixed tolerances on Brier and ECE reject 11% of identical candidates and, with recall's band, cap the gate's power at 81% (#134); a model with no measured noise floor still promotes on any positive margin (#137).

INJECTION-RECOVERY

Testing the model on answers we already know

Labels from archives are few, noisy and confounded. So the project also plants synthetic transits of known strength into real light curves and asks: does the model's score respond?



990 synthetic box transits: 40 host stars (24 false-positive hosts, 16 planet hosts), up to three periods each (110 host–period cells), nine signal-to-noise levels including zero. "Recovered" means a calibrated score of at least 0.486, the model's fitted threshold. Each point is the extra recovery rate over the same host with nothing injected; bars are 95% intervals.

The design

A **blocked factorial**: host star is the block, signal-to-noise and period are the treatments, and S/N = 0 is the control. Each host is compared with itself, which removes the star's own baseline from the comparison:

$$\mathrm{lift}_h(S) = r_h(S) - r_h(0), \quad t = \overline{\mathrm{lift}} \,/\, (s_{\mathrm{lift}} / \sqrt{40})$$

39 df

Injected depth is set from the host's own noise so each level means the same thing on every star:

$$\delta = \mathrm{S/N} \cdot \mathrm{CDPP} / \sqrt{N_{\mathrm{tr}}}$$ the standard transit S/N approximation

What it found

- **No measurable response below S/N 10.** At S/N 7 the recovery rate moves by 6 points, inside the noise; at S/N 10 by 20 (p = 0.003). Half the achievable lift is reached at S/N ≈ 14.
- **The score partly reads the star.** With nothing injected, 26% of trials still clear the threshold: 12% on false-positive hosts, 47% on planet hosts.

Why this is the metric worth reporting. It is measured on transits whose answer is known, it can be compared with published completeness curves, and it measures directly the faint-signal regime a discovery search would live in. The plan is to scale it from 990 to about 9,600 injections and model the curve with a floor term and host-clustered errors, rather than binning it (#97, #116).

This analysis re-reads existing injection runs and was not pre-registered; that is stated in its record. The observing-history confound (page 5) is the other reason label-based metrics are not enough on their own.

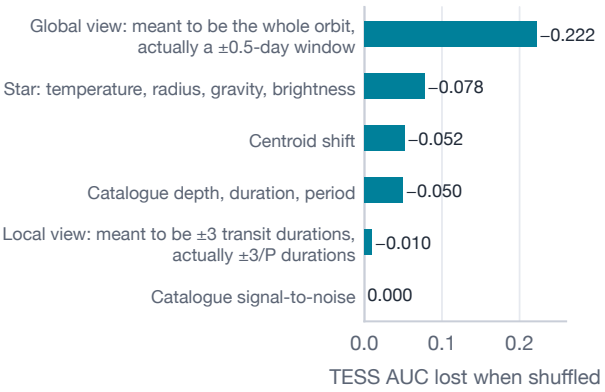
WHERE IT STANDS

An independent check: what the model really relies on

Eight challengers were measured against the served model and none replaced it. In September 2026 a simple check asked the question that had been skipped: does the network beat anything simple?

MODEL, SAME STARS AND FOLDS AS THE GATE	TESS AUC	TESS RECALL@1%	KEPLER AUC	KEPLER RECALL@1%
Served CNN (views + 9 numbers)	0.910 0.899–0.922	0.307 0.222–0.361	**0.991** 0.989–0.994	**0.799** 0.742–0.875
Gradient-boosted trees, 13 vetting numbers	**0.933** 0.924–0.943	**0.326** 0.231–0.458	0.988 0.984–0.991	0.735 0.676–0.811
Gradient-boosted trees, the CNN's 9 numbers	0.904 0.892–0.916	0.171 0.096–0.279	0.985 0.981–0.989	0.754 0.636–0.823
13-number trees minus CNN, paired	+0.023 +0.013 to +0.033	+0.019 −0.064 to +0.168	−0.004 −0.007 to −0.001	−0.063 −0.157 to +0.018

One configuration fixed before any result was seen, never tuned; trained per fold on training rows only. Differences are point estimates with 95% intervals from 2,000 paired bootstrap resamples; recall at 1% FPR is a step statistic, so its intervals are skewed rather than centred on the estimate. The bootstrap covers only which stars were sampled; the CNN's retraining noise is separate, and +0.023 also clears the floor it implies (0.013 to 0.019, depending on the seed estimate). Exploratory: it will be repeated under the project's own pre-registration (#104) before anything rests on it.



Each fold's model, re-scored with one input shuffled across stars. The network leans on the ±0.5-day window; shuffling the view meant to carry the transit's shape costs only 0.010.

Reading it

- On TESS, trees given only the nine numbers score close to the network with its light-curve views (0.904 against 0.910), and trees given the four extra vetting numbers score higher (0.933). Those four carry evidence the served views drop.
- One reason the champion survived: every contest was one challenger against the champion, with no simple baseline alongside. One challenger that did better, an ensemble keeping 43.6% of TESS planets against 30.5% for a single dual-view model scored on the same folds, was kept as a complement rather than served, because serving it would also mean deploying the branch model, work deferred when that line closed.
- Much of the machinery works: the check was quick to run because the out-of-fold predictions are saved.

ESTABLISHED TOOL	WHAT IT LOOKS AT	BUILT AND TESTED ON
Robovetter (Thompson 2018)	Hand-set tests on pipeline diagnostics: odd/even, secondary, centroid, shape	All Kepler DR25 signals; tuned with injected, inverted and scrambled light curves
ExoMiner++ (Valizadegan 2025)	Whole-orbit flux, odd/even, secondary, centroid, difference image, periodogram	57,162 labelled TESS signals; recall 0.951 at a 0.47% false-positive rate on its own test set, so at least 0.951 at 1%, against 0.307 here
TRICERATOPS (Giacalone 2021)	Bayesian false-positive probability with Gaia neighbours; not a classifier	No labels needed; this project runs a patched copy for validation

What is different here: the served inputs lack much of the evidence those tools use (partly the fixable window bug, partly diagnostics not yet built, such as difference images); the data is 2,367 TESS stars against 57,162 signals (a design choice: one row per star); and the populations differ, because this project's TESS negatives had already passed initial vetting to become TOIs; which way that cuts has not been established.

ENGINEERING

How the system is built and run

About 18,000 lines of Python (tests not counted) in a library and a scoring API, plus a static web console and the operational glue that keeps the data current.

Repository

```
pipeline/src/exoplanet_hunter/
  data/         catalogue, downloads, Gaia, ExoFOP
  validation/   gates that raise; promotion gate
  preprocess/   clean, detrend, fold, views
  features/     aux vector, centroid, pink S/N
  datasets/     TFRecords, tf.data, aux transform
  models/       dual-view CNN, MC dropout
  training/     cross-validation, calibration
  eval/         metrics, injection-recovery
  scoring/      the serving path, end to end
  search/       box-least-squares (discovery)
api/app/        FastAPI routes and schemas
frontend/design-console/   the web console
docs/           report, plan, limits, record
```

Tools

Python 3.11 · TensorFlow 2.17 / Keras · lightkurve 2.6 · scikit-learn · pandas · Hydra configs · **MLflow** run tracking · **DVC** data versioning on Cloudflare R2 · **Prefect** weekly refresh (launchd on the laptop) · **FastAPI** on Fly.io, scale-to-zero · console on Render · GitHub Actions CI.

Using the API

```
# the ranked catalogue, best first
API=https://exoplanet-hunter-api.fly.dev
curl -G "$API/candidates" \
  -d disposition=PC -d sort_by=prob_mean \
  -d order=desc -d limit=1

{ "name": "TOI-4328.01", "tic_id": 77175217,
  "period_days": 703.79, "depth_ppm": 601.96,
  "prob_mean": 0.955, "prob_std": 0.113,
  "prob_p10": 0.910, "prob_p90": 0.999,
  "fold_disagree": 0.026,
  "mc_disagree": 0.098,
  "score_source": "bulk-ensemble-mean" }

# score any star end to end: fetches its light
# curve, so a cold target can take minutes
curl "$API/score/341729521"
```

A real response from the live service, trimmed. The top candidate is itself a caution: two transits, a period ExoFOP's notes call likely too long, and S/N 9, below where page 12 finds any response to a transit. Other routes: `/model` (the model card), `/runs` (every recorded run), `/reliability`, `/healthz`.

Testing and checks

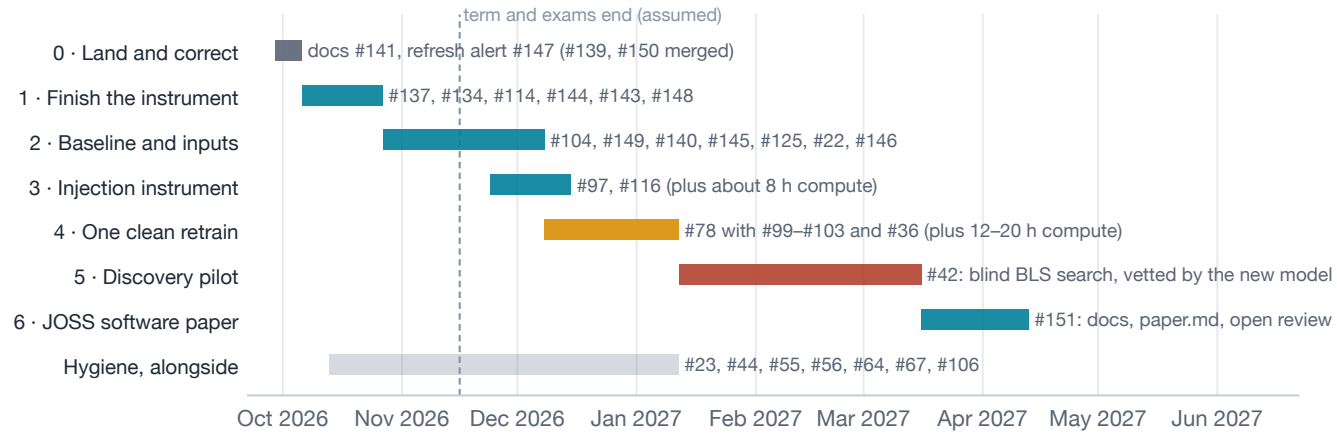
- **Unit and integration tests** for the pipeline and the API run in CI, with linting and the console build.
- **Validation gates** stop a data refresh on schema, coverage or leakage failures, rather than warning and carrying on.
- `make ready`, a 15-check "fit to show" gate: docs, links, tests, types, console build, refresh health.
- **Missing, and next (#146):** checks that fail when the science is wrong: inputs match their specification, the model beats a baseline, its score rises with transit strength, training and serving get identical inputs.

Operational rules. The weekly refresh pulls new labels, re-runs the gates and retrains only when 25 or more new labels arrive, but it never promotes a model: promotion is always a person's decision, with the gate's evidence in front of them. Deploying the API is a manual step. Data version bumps are separate commits from code changes, so a result can be traced to both.

WHAT IS LEFT

The plan, and a realistic timeline

Each item below is tracked on GitHub, with an estimate. The order follows the measurements: fix what the model sees and put a baseline beside it before the clean retrain, or the retrain inherits both problems.



Projection, not a commitment. It assumes about 8 hours a week through term and exams, then 15 to 20 hours a week over the summer; compute runs overnight on a laptop. Effort totals: phases 0 to 4, about 110 to 150 hours of work plus 30 to 50 hours of compute; the discovery pilot, another 40 to 80; the JOSS paper, 20 to 30 more plus the review.

PHASE	WORK (ISSUES)	EFFORT	OUTCOME
0 · Land and correct	Done: the noise-floor fix (#139) and the corrected console About page (#150). Still to do: correct the report (#141) and alert when the weekly refresh fails (#147)	3–5 h	The public story matches the code
1 · Finish the instrument	What a model with no noise floor means (#137); measured Brier/ECE floors (#134); a five-model champion baseline (#114); one recall definition (#144); training/serving input parity (#143); Model page floor (#148)	20–28 h	A gate sized to its nominal error rates
2 · Baseline and inputs	Pre-registered classical baseline (#104) and the decision it forces (#149); fix the view windows (#140), refit drifted epochs (#145), handle gaps and empty bins (#125) drawing on the data-driven gap correction of Wrench & Parashar (2025), the centroid flag (#22); one research check command (#146)	45–60 h	Inputs that carry the physics; a bar to beat
3 · Injection instrument	Model the completeness curve with a floor and host-clustered errors, scale to ~9,600 injections, place S/N levels where the curve moves (#97, #116)	10–14 h	The headline metric: completeness above the null floor
4 · One clean retrain	Corrected views, all 13 numbers, five models per fold, the Kepler cap lifted (#78, #36), with arms for ensembling, seed variance, AdamW, Kepler-to-TESS transfer and auxiliary heads (#99–#103)	30–40 h	A champion that beats the baseline under the gate, or a clear negative
5 · Discovery pilot	A job queue for blind box-least-squares search over TESS, ranked by the new model (#42)	40–80 h	A first shortlist of new candidates
6 · JOSS software paper	Document and package the pipeline, write a short paper on what it adds beside ExoMiner++, LEO-Vetter and TRICERATOPS, archive a release with a DOI, and submit to the Journal of Open Source Software for open peer review (#151, with #43)	20–30 h	A citable, peer-reviewed software paper
Alongside	Engineering hygiene: slow tests (#23), unowned risks (#44), typing (#55), data backup (#56), library consolidation (#64), test isolation (#67), cleanup audit (#106)	30–40 h	A smaller, safer codebase
Deferred	One row per signal instead of per star (#37, ~40 h); re-runs and re-tunes after the data changes (#38, #39); test-time augmentation (#117); upload and DOI features (#43); closed branch-line work (#40, #41)	90+ h	After the pilot, if it earns them

REFERENCE

Glossary and references

The terms this document leans on, and the work it builds on.

Transit: a planet passing in front of its star, dimming it slightly.

Light curve: a star's brightness measured over time.

TOI / KOI: a TESS or Kepler Object of Interest, a signal flagged for follow-up.

Phase-fold: wrapping time at the orbital period so every transit stacks on the others.

Eclipsing binary: two stars eclipsing each other, a common impostor.

Secondary eclipse: a second eclipse later in the orbit. A companion star often shows one; a planet's is tiny.

Blend: an eclipsing binary on a nearby star whose light leaks into the target's pixels.

Centroid: the centre of the light on the detector. If it moves during the dip, the dip is probably on another star.

CDPP: a star's noise level over the length of a transit.

Signal-to-noise (S/N): how far a dip stands above the noise.

ROC-AUC: the chance that a random planet outscores a random false positive.

Recall at 1% FPR: the share of real planets kept when only 1% of false positives get through.

Calibration (Brier, ECE): whether a score of 0.9 really means nine in ten.

Out of fold: scored by a model that never trained on that row.

Noise floor: the smallest difference a comparison can resolve, set by retraining with different random seeds.

MDE: minimum detectable effect, the smallest real improvement an AUC-only test catches 80% of the time.

Pre-registration: writing down how a result will be read before it exists.

Injection-recovery: planting synthetic transits of known strength and checking the model responds.

BLS: box-least-squares, a search for periodic box-shaped dips.

JOSS: the Journal of Open Source Software, a peer-reviewed journal for research software.

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Where to look

Code and issues: github.com/OliverDonaldson/exoplanet-hunter-v2 (public, MIT licence)

Full report: `docs/report.pdf` ; status and known weaknesses in `docs/PLAN.md` and `docs/known-limits.md`

Live: exoplanet-hunter-console.onrender.com · exoplanet-hunter-api.fly.dev

This work uses data from the NASA Exoplanet Archive and ExoFOP, operated by the NASA Exoplanet Science Institute; TESS and Kepler light curves from the Mikulski Archive for Space Telescopes (MAST); and stellar parameters from ESA's Gaia mission. Feedback is welcome as a GitHub issue.